

Beyond 4/4

A Fairness Study of Time Signature Control
in Text-to-Music Generation

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What is a Time Signature?

4/4

*"Twinkle Twinkle
Little Star"*

Count: 1-2-3-4

Most common in
Western pop

3/4

*"We Wish You A
Merry Christmas"*

Count: 1-2-3

The waltz
time signature



5/4

*"Mission: Impossible
Theme"*

Count: 1-2-3-4-5

Rare in western pop,
models often fail here



**Models easily fail,
tend to collapse to 4/4**



*Time signature defines how beats are grouped in each repeating cycle. (e.g. 4/4 means 4 beats per bar)
It isn't just a music theory concept -> when model default to 4/4, they marginalize musical cultures rely on other time signatures*

Research Questions

RQ1 User-Experience Bias

Do models handle cultural prompts (e.g., "waltz", "a song like Take Five") differently from formal prompts (e.g., "3/4 time", "5/4 time")?

RQ2 Cultural Bias

Do models systematically collapse non-4/4 time signatures into 4/4, under-representing musical traditions that rely on other time signatures?

Methodology

Models

Suno v4.5

State-of-the-art commercial model

Lyria 3 (Google)

Integrated into Gemini; first major LLM
with built-in text-to-music

Prompt Design

5 time signatures: 2/4, 3/4, 4/4, 5/4, 6/8

5 prompt pairs per time signature

Cultural: uses common musical language
(genre/tradition terms).

Formal: explicit time signature notation

Meter	Cultural Prompt	Formal Prompt
2/4	"An energetic march with brass and snare drums."	"An energetic piece in 2/4 time with brass and snare drums."
3/4	"A dramatic waltz with full orchestra and sweeping strings."	"A dramatic piece in 3/4 time with full orchestra and sweeping strings."

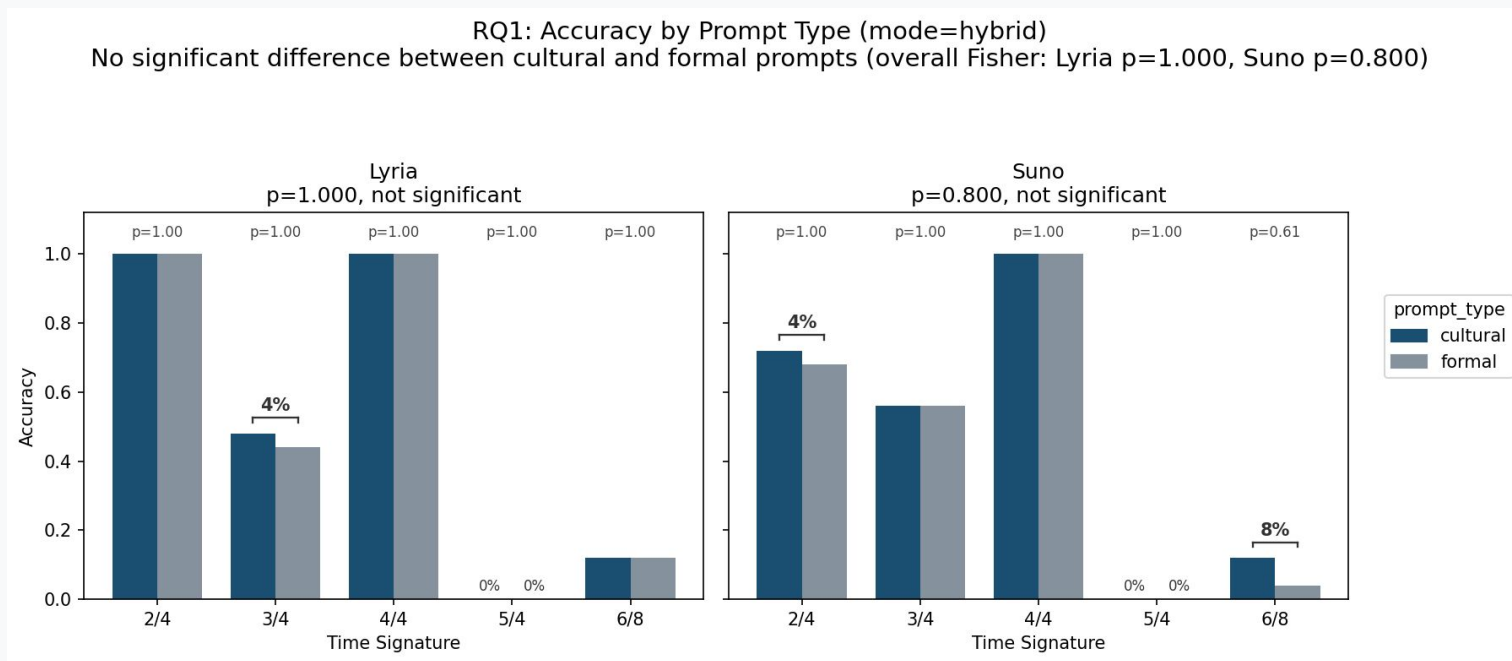
how to judge generation result: Hybrid Approach

Automatic time signature detection via madmom (MIR library) → Manual review for confidence < 0.9

250 songs × 2 models = 500 total generations

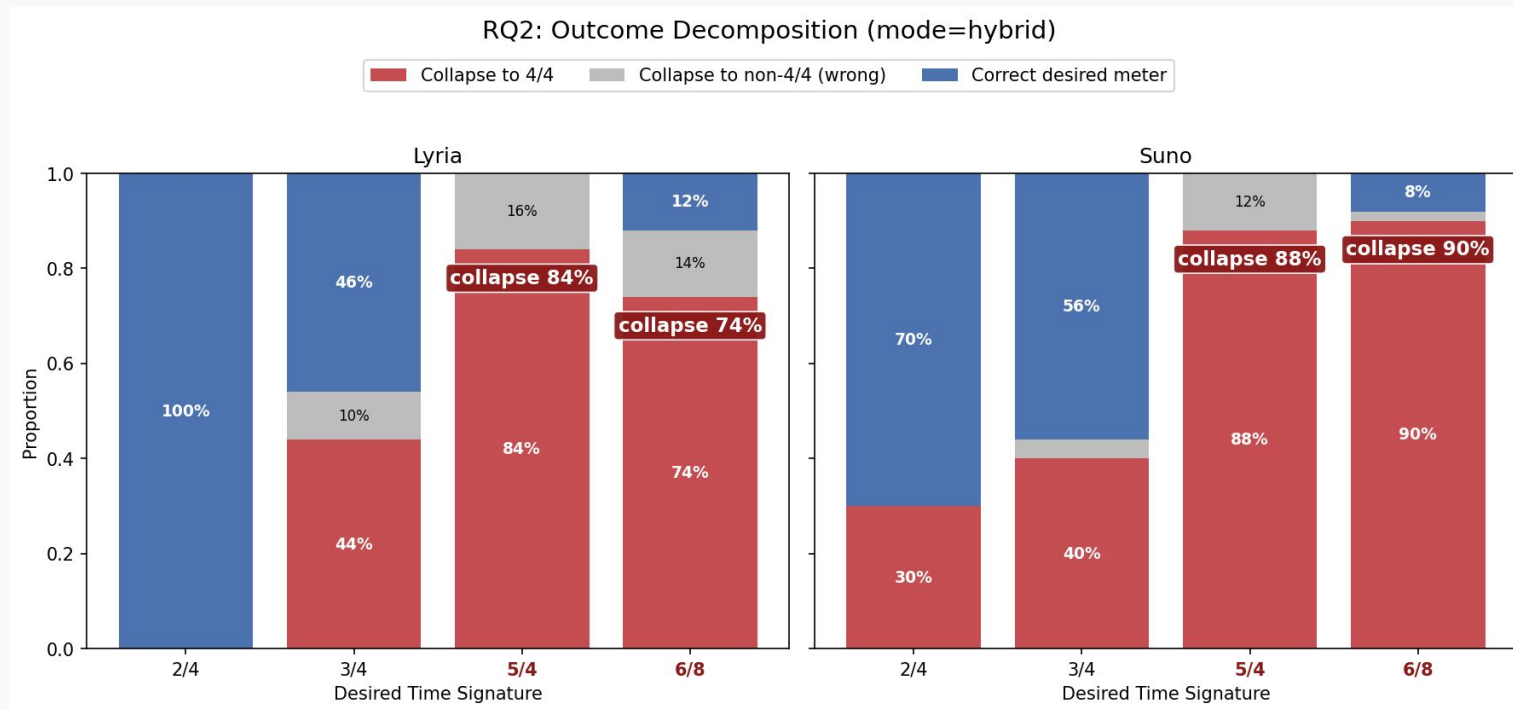
RQ1: Prompt Type Does Not Matter

No statistically significant difference between cultural and formal prompts (Fisher's exact test)



Finding: Both models treat cultural and formal prompts equally. User-experience bias is not supported by the data.

RQ2: Strong Collapse to 4/4 for 5/4 and 6/8



Finding: 5/4 and 6/8 collapse to 4/4 at 74–90% rate.

time signatures other than 4/4 are underrepresented, and many outputs tend to collapse to 4/4.

Discussion: Fairness Implications

Marginalized traditions when models default to 4/4:

5/4 Jazz

Take Five, Dave Brubeck Quartet

6/8 Irish

Jigs — compound meter is core identity

7/8 Balkan

Asymmetric groupings, folk traditions

User-level consequences:

Communities where non-4/4 is common

receive lower-quality service for culturally meaningful requests.

Musicians & producers needing specific meters face extra prompting effort and editing.

Bias is not about prompt phrasing — it reflects the model's underlying training distribution.

Conclusion

RQ1

No significant difference between cultural and formal prompts — user-experience bias not found

RQ2

5/4 and 6/8 collapse to 4/4 at 74–90% — strong evidence of cultural bias

Current text-to-music models provide uneven rhythmic coverage.

Improving robustness on non-4/4 time signatures is important for both quality and fairness.